

Qatar-1b

R-2835

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_150819 · created 2026-09-06T00:03:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457253.68134 +/- 0.00062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.016 +/- 0.053
Transit depth (Rp/Rs)^2	0.026 +/- 0.17 [%]
Orbital Inclination (inc)	79.98 +/- 2.91
Airmass coefficient 1 (a1)	0.177 +/- 0.1
Airmass coefficient 2 (a2)	1.29 +/- 1.37
Scatter in the residuals of the lightcurve fit is	50.42 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	4.5
Transit Duration (day)	0.02 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.016 $\pm$ 0.053 vs 0.1463 (deviation 1.61 σ)
Duration fit vs published	0.02 d vs 0.0692 d (deviation 1.35 σ)
Mid-time O-C	1.0 min
Depth SNR	0.2
Residual scatter	50.42 %

Light curve

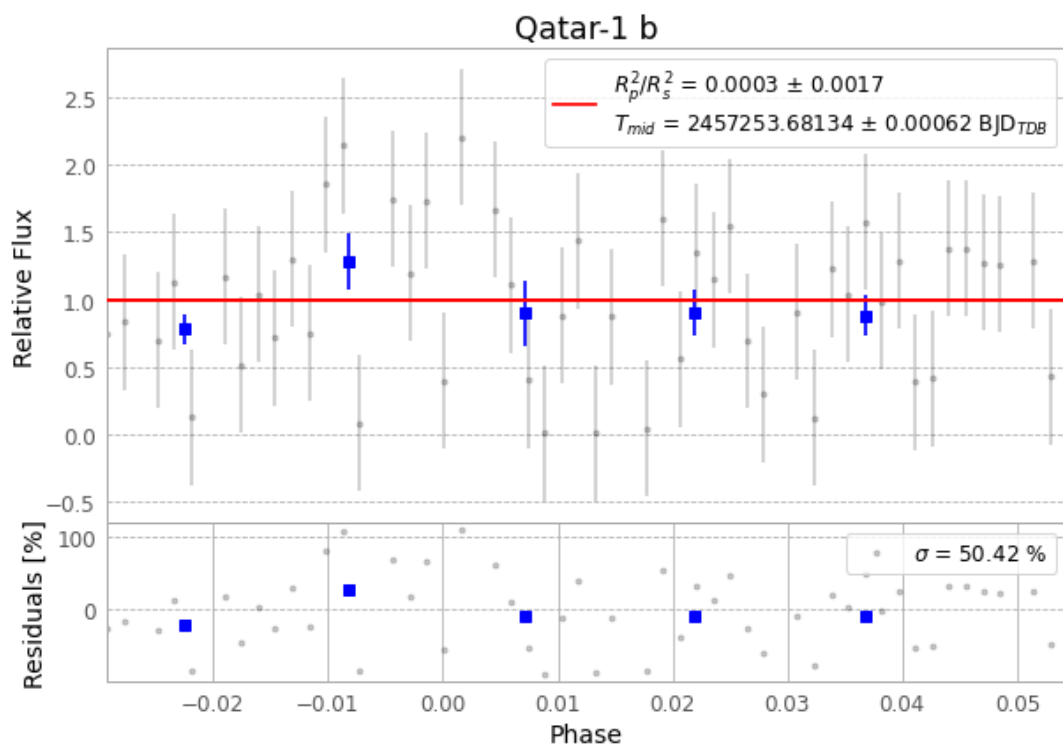


Figure 1 — FinalLightCurve_Qatar-1 b_2015-08-18.png (SHA-256 a5203b6e553c5233...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [455, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 75312c558b8d5587...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

59 FITS frames (39 MB), Qatar-1150819031512.FITS → Qatar-1150819060912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150819.log (c0bb3aefadfd...), FinalLightCurve_Qatar-1 b_2015-08-18.png (a5203b6e553c...), FinalLightCurve_Qatar-1 b_2015-08-18.csv (5ffc8f1d9ded...)

Compute resources

Wall time 182.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-06 00:03Z.